Functional Programming in Python

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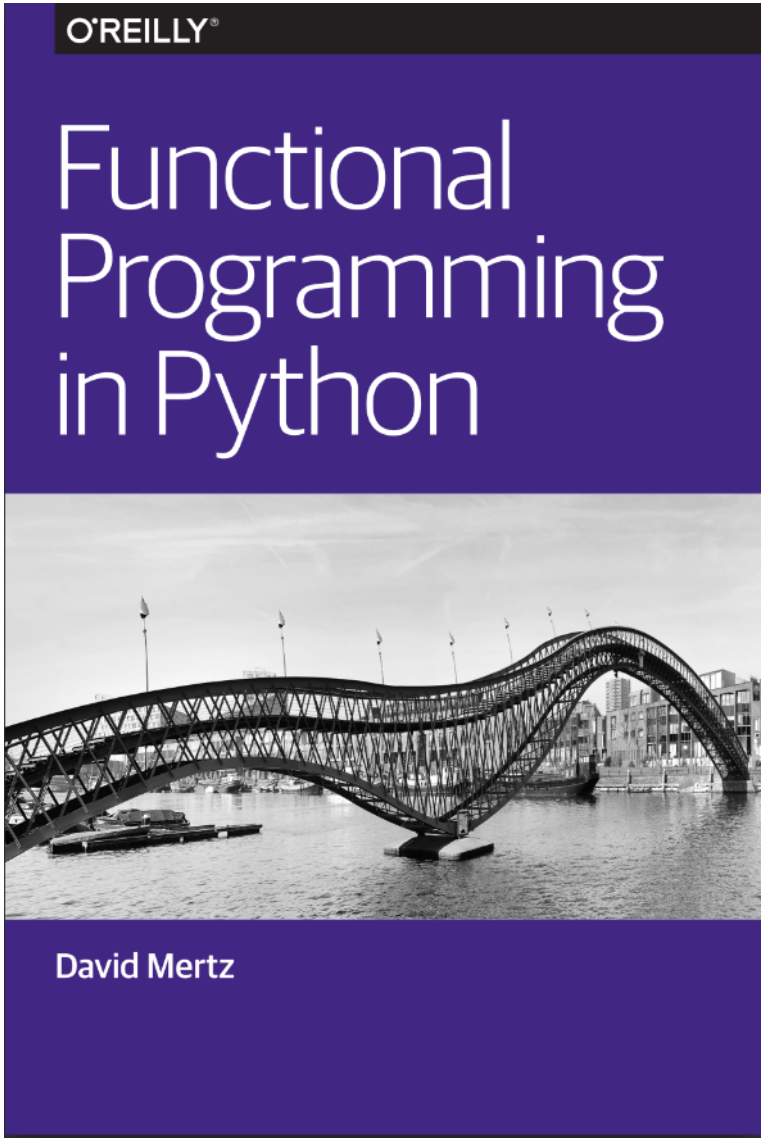
Based on



Functional Programming in Python

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Hints

Type hints improve code readability and help with static type checking.

```
>>> from typing import List, Dict, Tuple

>>> hello : str = "Hello, World!"
>>> names : List[str] = ["Alice", "Bob"]
>>> scores : Dict[str, int] = {"Alice": 100, "Bob": 95}
>>> vector : Tuple[int, int, int] = (1, 2, 3)
```

They can be also used in function definitions:

```
>>> def say_hello(names: List[str]) -> int:
...     for name in names:
...         print(f"Hello, {name}!")
...     return len(names)
```

Hints are stored in the `__annotations__` attribute of the function.

```
>>> say_hello.__annotations__
{'names': List[str], 'return': int}
```

Type Aliases

Type aliases can make complex type hints more readable and easier to manage. Using the `TypeAlias` from the `typing` module:

```
>>> from typing import TypeAlias, Tuple

>>> Vector3D : TypeAlias = Tuple[float, float, float]
```

Alternatively, using the `type` keyword:

```
>>> type Vector3D = Tuple[float, float, float]
```

The `Vector3D` alias represents a 3D vector as a tuple of three floats.

```
>>> v1 : Vector3D = (1.0, 2.4, 3.0)
>>> v2 : Vector3D = (4.0, 5.0, 6.1)
>>> add(v1, v2)
(5.0, 7.4, 9.1)
```

namedtuple

collections module provides the namedtuple function to create simple, immutable record types.

```
>>> from collections import namedtuple
>>> Point = namedtuple('Point', ['x', 'y'])
>>> p = Point(1, 2)
```

Instances of Point have named fields and can be accessed by index or attribute.

```
>>> p.x
1
>>> p[1]
2
>>> p
Point(x=1, y=2)
```

dataclass

dataclasses module provides a dataclass decorator to create classes with less boilerplate.

```
>>> from dataclasses import dataclass

>>> @dataclass(frozen=True)
... class Point:
...     x: int
...     y: int
```

Instances of Point are mostly immutable by setting the frozen parameter to True.

```
>>> p = Point(1, 2)
>>> p.x
1
>>> p.y
2
>>> p
Point(x=1, y=2)
```

List Comprehensions

List comprehensions provide a concise way to create lists.
Traditional `for` loop to filter and modify elements in a list:

```
>>> collection = []
>>> for datum in data_set:
...     if condition(datum):
...         collection.append(datum)
...     else:
...         new = modify(datum)
...         collection.append(new)
```

Using a list comprehension for the same task:

```
>>> collection = [datum if condition(datum) else modify(datum) for
    datum in data_set]
```

List comprehensions are more compact, readable, and often faster than traditional loops.

Other Comprehensions

Dictionary comprehension to create a mapping of indices to characters:

```
>>> {i:chr(65+i) for i in range(6)}
{0: 'A', 1: 'B', 2: 'C', 3: 'D', 4: 'E', 5: 'F'}
```

Set comprehension to create a set of characters:

```
>>> {chr(65+i) for i in range(6)}
{'A', 'B', 'C', 'D', 'E', 'F'}
```

Generator expression to lazily generate characters:

```
>>> (chr(65+i) for i in range(6))
<generator object <genexpr> at 0x7f8b1c1b3d60>
```

Generators

Generator functions declare a function that behaves like an iterator, allowing it to be used in a for loop. Unlike lists, they are lazy and do not store all the values in memory.

```
>>> def count(n):
...     for i in range(n):
...         yield i

>>> c = count(5)
>>> c
<generator object count at 0x7f8b1c1b3d60>
>>> list(c)
[0, 1, 2, 3, 4]
```

Recursion

Python supports recursion, but it is not optimized for tail calls or deep recursion. It is best suited for problems that can be solved with a small number of recursive calls.

```
>>> def factorialI(N):
...     product = 1
...     while N >= 1:
...         product *= N
...         N -= 1
...     return product


>>> def factorialR(N):
...     return 1 if N <= 1 else N * factorialR(N-1)
```

Pure Functions

Pure functions are functions where the output value is determined only by its input values, without observable side effects.
An impure function example:

```
>>> def print_and_add(x, y):
...     print(f"Adding {x} and {y}")
...     return x + y

>>> print_and_add(5, 3)
Adding 5 and 3
8
```

The function `print_and_add` is impure because it has a side effect (printing).

```
>>> def add(x, y):
...     return x + y

>>> add(5, 3)
8
```

Lambda Functions

Lambda functions are small anonymous functions defined with the `lambda` keyword and they can be used wherever function objects are required.

```
>>> double = lambda x: x * 2
>>> double(3)
6
```

The name of a lambda function can be accessed or set using the `__qualname__` attribute.

```
>>> double.__qualname__
'lambda'
>>> len.__qualname__
'len'
```

Closures

Python supports closures - functions that refer to variables from the scope in which they were defined.

```
>>> def f(x):
...     def g(y):
...         return x + y
...     return g

>>> add3 = f(3)
>>> add3(3)
6
>>> add3(5)
8
```

Unexpected Behavior

Closures can lead to unexpected behavior when the enclosed variable is mutable.

```
>>> adders = []
>>> for n in range(5):
...     adders.append(lambda m: m+n)
>>> [adder(10) for adder in adders]
[14, 14, 14, 14, 14]
>>> n = 10
>>> [adder(10) for adder in adders]
[20, 20, 20, 20, 20]
```

Unexpected Behavior

Fortunately, a small change brings behavior that probably better meets our goal:

```
>>> adders = []
>>> for n in range(5):
...     adders.append(lambda m, n=n: m+n)
>>> [adder(10) for adder in adders]
[10, 11, 12, 13, 14]
>>> n = 10
>>> [adder(10) for adder in adders]
[10, 11, 12, 13, 14]
>>> add4 = adders[4]
>>> add4(10, 100)  # Can override the bound value
110
```

Callable Instances

Callable instances can be created in Python by defining the `__call__` method in a class. This approach is useful for creating objects that behave like functions.

```
>>> class Adder:
...     def __init__(self, x):
...         self.__x = x
...     def __call__(self, y):
...         return self.__x + y
```

`Adder` instances can be called with an argument to add a predefined value.

```
>>> add5 = Adder(5)
>>> add5(10)
15
>>> add5(3)
8
```

match Statement

```
>>> from dataclasses import dataclass
>>> from typing import Union
```

Define dataclasses for shapes:

```
>>> @dataclass
... class Circle:
...     radius: float

>>> @dataclass
... class Rectangle:
...     width: float
...     height: float

>>> @dataclass
... class Triangle:
...     base: float
...     height: float
```

A union type to represent all shapes:

```
>>> Shape = Union[Circle, Rectangle, Triangle]
```

match Statement

match can be used to define a function that describes a shape:

```
>>> def describe_shape(shape: Shape) -> str:
...     match shape:
...         case Circle(radius=r):
...             return f"A circle with radius {r}."
...         case Rectangle(width=w, height=h):
...             return f"A rectangle with width {w} and height {h}.
    "
...         case Triangle(base=b, height=h):
...             return f"A triangle with base {b} and height {h}."
...         case _:
...             return "Not a valid shape."

>>> describe_shape(Circle(5))
'A circle with radius 5.'
>>> describe_shape(Rectangle(10, 20))
'A rectangle with width 10 and height 20.'
>>> describe_shape(Triangle(8, 6))
'A triangle with base 8 and height 6.'
>>> describe_shape(5)
'Not a valid shape.'
```

match Statement for Lists

match statement can also be used to define a function that recursively sums a list of numbers:

```
>>> def sum_list(numbers):
...     match numbers:
...         case []:
...             return 0
...         case [head, *tail]:
...             return head + sum_list(tail)

>>> numbers = [1, 2, 3, 4, 5]
>>> sum_list(numbers)
15
>>> sum_list([])
0
```

Higher-Order Functions

In Python, functions are first-class objects, allowing them to be passed as arguments, returned from other functions, and assigned to variables.

```
>>> def shout(text):
...     return text.upper()
...
>>> def whisper(text):
...     return text.lower()
...
>>> def greet(func):
...     return func("Hi, I am created by a function passed as an
    argument.")

>>> greet(shout)
'HI, I AM CREATED BY A FUNCTION PASSED AS AN ARGUMENT.'
>>> greet(whisper)
'hi, i am created by a function passed as an argument.'
```

Function Composition

Using higher-order functions, it is possible to compose functions by chaining them together and creating a simple pipeline.

```
>>> def compose(*funcs):
...     def inner(data, funcs=funcs):
...         result = data
...         for f in reversed(funcs):
...             result = f(result)
...         return result
...     return inner

>>> times2 = lambda x: x*2
>>> minus3 = lambda x: x-3
>>> mod6 = lambda x: x%6
>>> f = compose(mod6, times2, minus3)
>>> all(f(i)==((i-3)*2)%6 for i in range(1000000))
True
```

Decorators

Decorators provide a convenient way to wrap additional behavior around function calls, allowing reusable modifications like logging or validation.

```
>>> def decorator(func):
...     def wrapper():
...         print("Before calling the function.")
...         func()
...         print("After calling the function.")
...     return wrapper

>>> @decorator
... def greet():
...     print("Hello, World!")

>>> greet()
Before calling the function.
Hello, World!
After calling the function.
```

The operator Module

This module provides function equivalents of Python's built-in operators, making it easier to pass them around or compose when building transformations alongside other utilities.

```
>>> import operator

>>> operator.add(1, 2) == 1 + 2
True
>>> operator.mul(3, 4) == 3 * 4
True
>>> operator.pow(2, 3) == 2 ** 3
True
```

The itertools Module

The itertools module provides iterable utilities such as accumulate for partial computations or repeated applications of a function.

```
>>> from operator import mul
>>> from itertools import accumulate, repeat
>>> data = [3, 4, 6, 2, 1, 9, 0, 7, 5, 8]

>>> list(accumulate(data, max))
[3, 4, 6, 6, 6, 9, 9, 9, 9, 9]

>>> list(accumulate(data, mul))
[3, 12, 72, 144, 144, 1296, 0, 0, 0, 0]

>>> update = lambda balance, payment: round(balance * 1.05) -
    payment
>>> list(accumulate(repeat(90, 10), update, initial=1000))
[1000, 960, 918, 874, 828, 779, 728, 674, 618, 559, 497]
```

The functools Module

The functools module provides higher-order functions that operate on other functions, making it easier to compose or partially apply functions.

```
>>> from functools import reduce
>>> from operator import mul

>>> def factorial(n):
...     return reduce(mul, range(1, n+1), 1)


>>> from functools import partial

>>> def multiply(x, y):
...     return x * y

>>> double = partial(multiply, 2)
>>> double(4)
8
```

About returns

Make your functions return something meaningful, typed, and safe!

`https://github.com/dry-python/returns`

Maybe

```
>>> @dataclass
... class Address(object):
...     street: Optional[str]

>>> @dataclass
... class User(object):
...     address: Optional[Address]

>>> @dataclass
... class Order(object):
...     user: Optional[User]

>>> def get_street_address(order: Order) -> Maybe[str]:
...     return Maybe.from_optional(order.user).bind_optional(
...         lambda user: user.address,
...     ).bind_optional(
...         lambda address: address.street,
...     )
```

Maybe

```
>>> with_address = Order(User(Address('Some street')))
>>> empty_user = Order(None)
>>> empty_address = Order(User(None))
>>> empty_street = Order(User(Address(None)))

>>> get_street_address(with_address)
<Some: Some street>

>>> get_street_address(empty_user)
<Nothing>

>>> get_street_address(empty_address)
<Nothing>

>>> get_street_address(empty_street)
<Nothing>
```

Maybe Decorator

```
>>> from returns.maybe import maybe

>>> @maybe
... def number(num: int) -> Optional[int]:
...     if num > 0:
...         return num
...     return None

>>> number(1)
<Some: 1>
>>> number(-1)
<Nothing>
```

Result

```
>>> from returns.result import Result, Success, Failure

>>> def find_user(user_id: int) -> Result['User', str]:
...     user = User.objects.filter(id=user_id)
...     if user.exists():
...         return Success(user[0])
...     return Failure('User was not found')

>>> find_user(1)
<Success(User{id: 1, ...})>

>>> find_user(0)
<Failure('User was not found')>
```

Result Decorator

```
>>> from returns.result import safe

>>> @safe  # Will convert type to: Callable[[int], Result[float,
    Exception]]
... def divide(number: int) -> float:
...     return number / number

>>> divide(1)
<Success(1.0)>

>>> divide(0)
<Failure: division by zero>
```

Pointfree

```
>>> from returns.pointfree import map_

>>> def as_int(arg: str) -> int:
...     return ord(arg)

>>> container : Maybe[str] = Some('a')
>>> map_(as_int)(container) == container.map(as_int)
True
```

Trampolines

```
>>> from typing import Union, List
>>> from returns.trampolines import Trampoline, trampoline

>>> @trampoline
... def accumulate(numbers: List[int], acc: int = 0) -> Union[int,
    Trampoline[int]]:
...     if not numbers:
...         return acc
...     number = numbers.pop()
...     return Trampoline(accumulate, numbers, acc + number)

>>> accumulate([1, 2])
3
>>> accumulate([1, 2, 3])
6
```

Fold

```
>>> from typing import Callable
>>> from returns.maybe import Some
>>> from returns.iterables import Fold

>>> def sum_two(first: int) -> Callable[[int], int]:
...     return lambda second: first + second

>>> Fold.loop(
...     [Some(1), Some(2), Some(3)],
...     Some(10),
...     sum_two,
... )
<Some: 16>
```

Pipelines

```
>>> from returns.result import Result, Success, Failure

>>> def regular_function(arg: int) -> float:
...     return float(arg)

>>> def returns_container(arg: float) -> Result[str, ValueError]:
...     if arg != 0:
...         return Success(str(arg))
...     return Failure(ValueError('Wrong arg'))

>>> def also_returns_container(arg: str) -> Result[str, ValueError
    ]:
...     return Success(arg + '!')
```

flow function

```
>>> from returns.pipeline import flow
>>> from returns.pointfree import bind

>>> flow(
...     1,
...     regular_function,
...     returns_container,
...     bind(also_returns_container),
... )
<Success: '1.0!'>

>>> flow(
...     0,
...     regular_function,
...     returns_container,
...     bind(also_returns_container),
... )
<Failure: Wrong arg>
```

pipe function

```
>>> from returns.pipeline import pipe
>>> from returns.pointfree import bind

>>> transaction = pipe(
...     regular_function,
...     returns_container,
...     bind(also_returns_container),
... )
>>> transaction(1)
<Success: '1.0!'>
>>> transaction(0)
<Failure: Wrong arg>
```

Mappable

From the `returns` documentation:
Something is considered mappable if we can map it using a function, generally map is a method that accepts a function.

```
>>> from returns.maybe import Maybe, Some

>>> def can_be_mapped(string: str) -> str:
...     return string + '!'

>>> maybe_str: Maybe[str] = Some('example')

>>> assert maybe_str.map(can_be_mapped) == Some('example!')
```

Applicative

From the `returns` documentation:
Something is considered applicative if it is a functor already and, moreover, we can apply another container to it and construct a new value with `.from_value` method.

```
>>> from returns.maybe import Maybe, Some

>>> maybe_str = Maybe.from_value('example')

>>> maybe_func = Maybe.from_value(len)  # we use function as a
    value!

>>> assert maybe_str.apply(maybe_func) == Some(7)
```

Bindable

From the `returns` documentation:
Bindable is something that we can bind with a function.

```
>>> from returns.maybe import Nothing, Maybe, Some

>>> def bindable(string: str) -> Maybe[str]:
...     return Some(string + 'b')

>>> assert Some('a').bind(bindable) == Some('ab')
>>> assert Nothing.bind(bindable) == Nothing
```

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Thank you!

Questions?